

Hierarchical Stability of GRA Obnullification: A Variational Formulation

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Abstract

A formalization of the GRA obnullification mechanism is proposed as a variational hierarchical functional that determines the dynamics of “foam” Φ in a multi-level system. The derivative $d\Phi/dh$ with respect to the level parameter h makes it possible to distinguish productive collapse of contradictions, stagnation, and degradation. Axioms, definitions, lemmas, and a theorem on the stability of obnullification are formulated, as well as a discrete algorithm with pseudocode for practical implementation.

1 Introduction

The GRA (Gradient Reduction of Argumentative Foam) framework aims to minimize “foam” Φ — a measure of contradictions, noise, and redundancy in hierarchical systems. The basic obnullification mechanism assumes that the top level of the hierarchy tends to $\Phi \approx 0$, which is interpreted as global coherence. In this work, obnullification is considered not only statically but also dynamically — through the derivative $d\Phi/dh$. This allows introducing a stability criterion for obnullification and investigating the system’s sensitivity to changes in the hierarchy.

2 Hierarchical Model and Foam Functional

A system with levels $l = 0, \dots, L$ is considered. The state of a level is $x_l \in \mathbb{R}^{n_l}$. The top level L sets global coherence. Local foam: $\Phi_l = \Phi(x_l) \geq 0$.

Hierarchical GRA-functional:

$$J[x_0, \dots, x_L] = \sum_{l=0}^L \alpha_l \Phi(x_l) + \sum_{l=0}^{L-1} \beta_l C_l(x_l, A_l x_{l+1}) + \gamma \Psi(x_L), \quad (1)$$

where:

- C_l — penalty for mismatch between adjacent levels,
- A_l — lifting/projection operator,
- $\Psi(x_L)$ — top-level obnullification condition,
- $\alpha_l, \beta_l, \gamma > 0$ — weights.

Under continuous parametrization of the level h , $\Phi(h)$ is differentiable, and $d\Phi/dh$ is the zero-gradient.

3 Axioms of GRA Obnullification

Axiom 1 (Hierarchy). *The system is organized into levels $l = 0, \dots, L$, each with its own state.*

Axiom 2 (Foam). *For each level a foam functional $\Phi(x_l) \geq 0$ is defined.*

Axiom 3 (Obnullification). *The operation T_l reduces foam: $\Phi(T_l(x_l)) \leq \Phi(x_l)$.*

Axiom 4 (Monotonicity). *In the ideal regime $\Phi_{l+1} \leq \Phi_l$.*

Axiom 5 (Sensitivity). *$\Phi(h)$ is differentiable, the sign of $d\Phi/dh$ determines the evolution mode.*

4 Definitions

Definition 1 (GRA-functional). *See Equation 1.*

Definition 2 (Stable obnullification). *At a point h^* : $\Phi(h^*) = 0$, $\frac{d\Phi}{dh}(h^*) = 0$, $\frac{d^2\Phi}{dh^2}(h^*) > 0$.*

Definition 3 (Discrete foam derivative). $\Delta\Phi_l = \Phi_{l+1} - \Phi_l$, *its discrete analogue of $d\Phi/dh$.*

Definition 4 (Zero-gradient). $\nabla_h\Phi := \frac{d\Phi}{dh}$, *characterizes the direction and rate of foam change along the hierarchy.*

5 Lemmas

Lemma 1 (Local stationarity). *At a minimum, the variational derivatives with respect to each x_l vanish.*

Lemma 2 (Zero foam does not guarantee stability). *It is also necessary that $\frac{d\Phi}{dh} = 0$.*

Lemma 3 (Positive derivative — degradation). *If $\frac{d\Phi}{dh} > 0$, foam grows.*

Lemma 4 (Discrete stability criterion). *For a level l^* : $\Phi_{l^*} = 0$, $\Phi_{l^*+1} - \Phi_{l^*} = 0$, $\Phi_{l^*+1} - 2\Phi_{l^*} + \Phi_{l^*-1} > 0$.*

6 Stability Theorem

Theorem 5. *Let $\Phi(h)$ be twice differentiable with respect to h and let there exist a point h^* such that*

$$\Phi(h^*) = 0, \quad \frac{d\Phi}{dh}(h^*) = 0, \quad \frac{d^2\Phi}{dh^2}(h^*) > 0. \quad (2)$$

Then obnullification at level h^ is stable: in some neighborhood perturbations do not lead to foam growth, and the system returns to $\Phi \approx 0$.*

7 Proof Sketch

1. From $\frac{d\Phi}{dh}(h^*) = 0$ it follows that h^* is a critical point.
2. The condition $\frac{d^2\Phi}{dh^2}(h^*) > 0$ means a local minimum.
3. There exists a neighborhood $U(h^*)$ where $\Phi(h) \geq \Phi(h^*) = 0$.
4. Since $\Phi \geq 0$, zero foam is locally minimal.
5. Under small perturbations a gradient process returns the system to h^* , which means stability. In the discrete case the reasoning is repeated via finite differences.

8 Discrete Algorithm and Pseudocode

The algorithm combines GRA obnullification, gradient alignment $d\Phi/dh$, and control of the second derivative.

Algorithm 1 HierarchicalStabilityOptimizer

```
1: class HierarchicalStabilityOptimizer
2:   def __init__(self, hierarchy, lr=0.01):
3:       self.h = hierarchy                                ▷ list of levels with .state and .foam()
4:       self.lr = lr
5:   def step(self):
6:       # 1. GRA obnullification
7:       for level in self.h: level.state = gra_nullify(level.state)
8:       # 2. Align dPhi/dh
9:       for l in range(len(self.h)-1):
10:          dPhi = self.h[l+1].foam() - self.h[l].foam()
11:          if abs(dPhi) > 1e-6:
12:              grad = grad_connection(self.h[l], self.h[l+1], dPhi)
13:              self.h[l+1].state -= self.lr * grad
14:       # 3. Ensure  $d^2\Phi/dh^2 > 0$ 
15:       for l in range(1, len(self.h)-1):
16:          d2Phi = self.h[l+1].foam() - 2*self.h[l].foam() + self.h[l-1].foam()
17:          if d2Phi <= 0:
18:              self.h[l].state += self.lr * 0.01 * (self.h[l+1].state + self.h[l-1].state - 2*self.h[l].state)
19:   def optimize(self, max_iter=100):
20:       for _ in range(max_iter): self.step()
```

The full implementation with abstract functions is located in `src/optimizer.py`.

9 Discussion and Applications

Interpretation of the derivative sign:

- $\frac{d\Phi}{dh} < 0$ — productive foam collapse
- $\frac{d\Phi}{dh} \approx 0$ — stagnation or plateau
- $\frac{d\Phi}{dh} > 0$ — degradation (hallucinations, growth of contradictions)

Applications:

- Diagnostics of cognitive stability of AI architectures (monitoring “foam” in hidden states of neural networks)
- Multi-level management in organizations (analyzing consistency of decisions across hierarchy levels)
- Analysis of the evolution of scientific theories (tracking contradictions in knowledge systems)

10 Installation and Usage

```
git clone https://github.com/qqewq/GRA-Hierarchical-Stability.git
cd GRA-Hierarchical-Stability
```

```
pip install -r requirements.txt
python examples/demo.py
```

To integrate with GRA-Multiverse-Final, replace the stubs in `src/foam_utils.py` with real functions from your project.

11 References

- GRA-Multiverse-Final
- GRA-Swarm
- NeurIPS 2022 hierarchical optimization materials
- HAL preprint hal-04088837
- SPP1962 preprint 036, WIAS Berlin